Soccer World Cup Prediction Through Statistical Models

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Introduction

Motivation

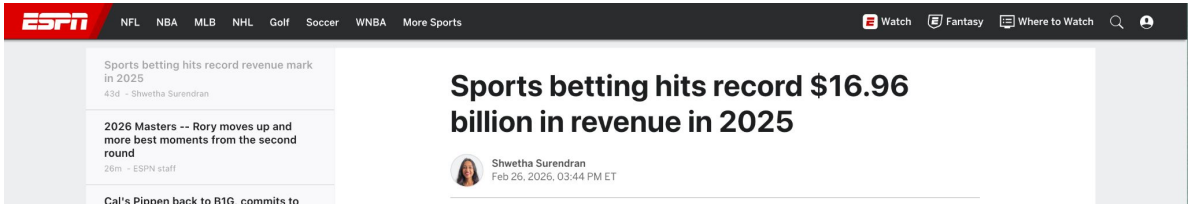
- Tournament Forecasting
- Performance Evaluation
- Seeding
- Valuation
- Strategy
- Recruitment
- Betting Market

FIFA anticipates $14 billion revenue for 2027-2030 cycle

By Reuters

March 19, 2026 2:22 PM EDT · Updated March 19, 2026

A drone view of the Banorte Stadium, also known as Azteca Stadium, the venue that will host the opening match of the 2026 FIFA World Cup and become the first stadium to hold three World Cups (1970, 1986, 2026), in Mexico City, Mexico, March 17, 2026. REUTERS/Luis Cortes/File Photo Purchase Licensing Rights



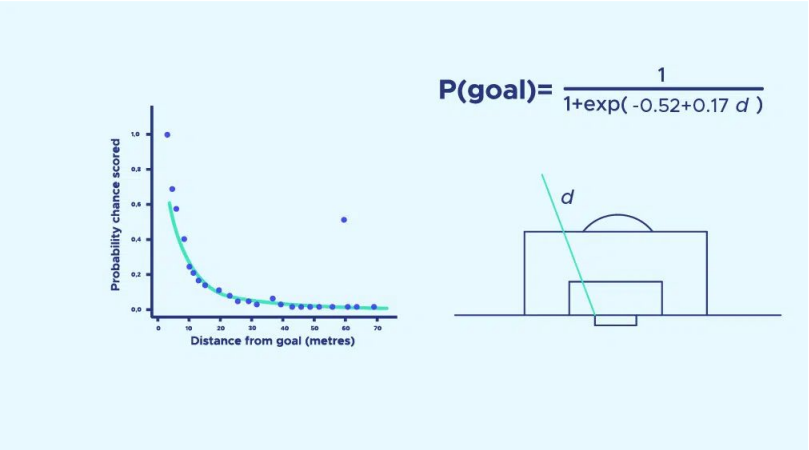
Project Overview

- Develop a head-to-head prediction model to forecast match outcomes (winner or draw) between international teams for the entire tournament.
  - Simulates the group stages with (W-L-D), and Knockouts with Win, Win AET, Win on Penalties.
- Scope: International teams participating in the 2026 FIFA World Cup.
- Integrating 5 years of international matchup history with 5 years of player statistics from Europe's top 5 domestic leagues.
  - (ENG) Premier League, (SPA) LaLiga, (ITA) Serie A, (GER) Bundesliga, (FRA) Ligue 1
- Models: Logistic regression using custom-engineered variables, SVM, RF, XGBoost.
- Important Metrics: ELO and xG.

$$E_{\mathsf{A}} = \frac{1}{1 + 10^{(R_{\mathsf{B}} - R_{\mathsf{A}})/400}} \ .$$

What exactly is xG?

- A statistic used from where a player shoots at goal, that measures the quality of the shot position in regard to the goal.
- Invented in 2012 from a dataset of over 1 million shots to create a normalized metric of probability of scoring.
- Used widely in team data analytics for expected performance as well as player fit transfers.
- xG features used: non-penalty xG, xG chain, and chance creation.



Questions

- **Player Impact:** How much do individual player stats from top European leagues influence their national team's win probability?
- **Data Synergy:** How effectively does combining 5-year international history with recent club-level data predict tournament outcomes?
- **Model Accuracy:** How accurately can a logistic regression model predict specific 2026 World Cup match outcomes (win/draw)?
- **Predicting Draws:** Which specific variables most strongly correlate with a match ending in a draw, given the sport's high variance?
- **World Cup Simulation:** How realistic is our prediction for the world cup winner?
- **Feature Impact:** Which player/team features matter most?
- **Group Stage:** Which teams maximize expected points in groups?
- **Bracket Modeling:** How do different models change the bracket?
- **Winning Forecast:** How sensitive is the champion forecast to model choice?

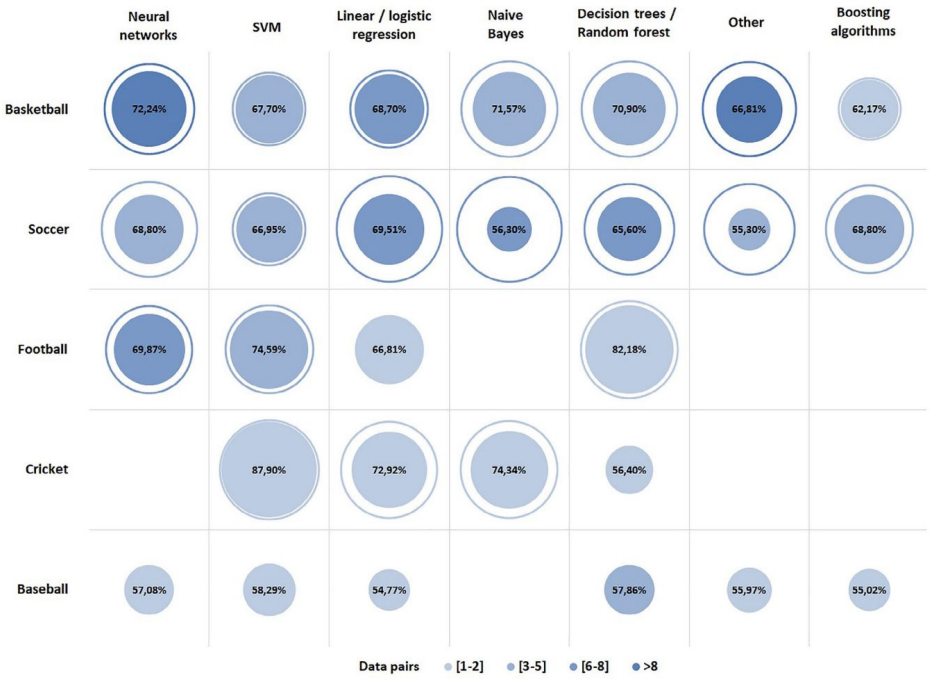
State of the Art

Club Level Metrics (Bat-Erdene and Peloguin (2025)).

- Advanced metrics in Europe's top five leagues are among the strongest predictors of a team's points per match.
- Crucial Offensive Variables: Expected goal difference (xGD) and chance creation (xG+xAG/90).
- Crucial Defensive Variables: Lower expected goals against (xGA) effectively separates elite squads from average ones.
- Validates our use of aggregated domestic expected metrics as standardized indicators of international squad quality.

ML in Sports (Horvat and Job (2020))

- Review of 39 papers across five major sports. Neural networks, Support Vector Machines, Naïve Bayes, Random Forests, and Logistic Regression.



ML Soccer Prediction (Bunker, Yeung, Fujii (2024))

- Top-Performing Models: Gradient-Boosted Trees (e.g., XGBoost, CatBoost) currently lead the field, particularly when using historical goal data.
- Bookmaker odds remains a formidable baseline; many sophisticated ML models still struggle to significantly outperform market prices.
- Deep Learning loses interpretability, one of our objectives with classical methods

Materials and Methods

Datasets

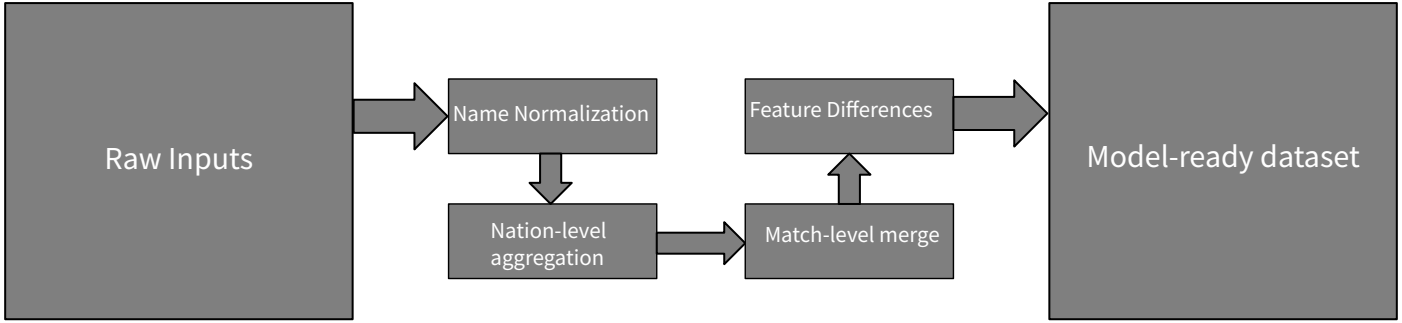
Data came from 4 different sources:

1. International Match Results (2021-2025) from `worldfootballR` package
   a. Their information is from Football statistics websites like FBref or Transfermarkt
2. International Teams ELO ratings
   a. https://www.kaggle.com/datasets/saifalnimri/international-football-elo-ratings
3. Scraped player data from 2021-2022 to 2025-2026 (up until March 31st) seasons for all players in Top 5 Leagues
   a. Statsbomb and understat data
4. Transfermarkt updated data to 2026
   a. https://www.kaggle.com/datasets/davidcariboo/player-scores/data

- Only Teams that have qualified for the 2026 World Cup were kept.
- Same with players that play for these teams.

1. Data Merging & Feature Construction

- Standardize team names across all sources
- Aggregate player-level stats into nation-level squad features
- Merge match history with team features and ELO
- Convert absolute stats into difference features for modeling
- **Symmetrical Data Augmentation:** Doubled the dataset size by creating a mirrored copy of every match where "Home" and "Away" labels (and their stats) are swapped.



2. Target Creation & Feature Engineering

- Aggregate player stats into national-team squad features.
  - xG/90, non-pen xG/90, xA/90, shots/90, key passes/90, xG chain, xG buildup, etc.
- Weight features by minutes played.
- Build depth and concentration measures from top players.
  - Attacking depth index, spread of contribution amongst players.
- Convert absolute team stats into home-away difference features.
  - Home - Away differences, Home + Away totals (xG_diff, np_xG_diff, xA_diff, shots_diff, key_passes_diff).
- Add ELO and match-context variables.
- Use these features to predict scorelines, draws, and knockout winners.
  - Group stage: home_win, draw, away_win or scoreline/expected points.
  - Knockout stage: winner after 90, then extra time, then penalties if needed.

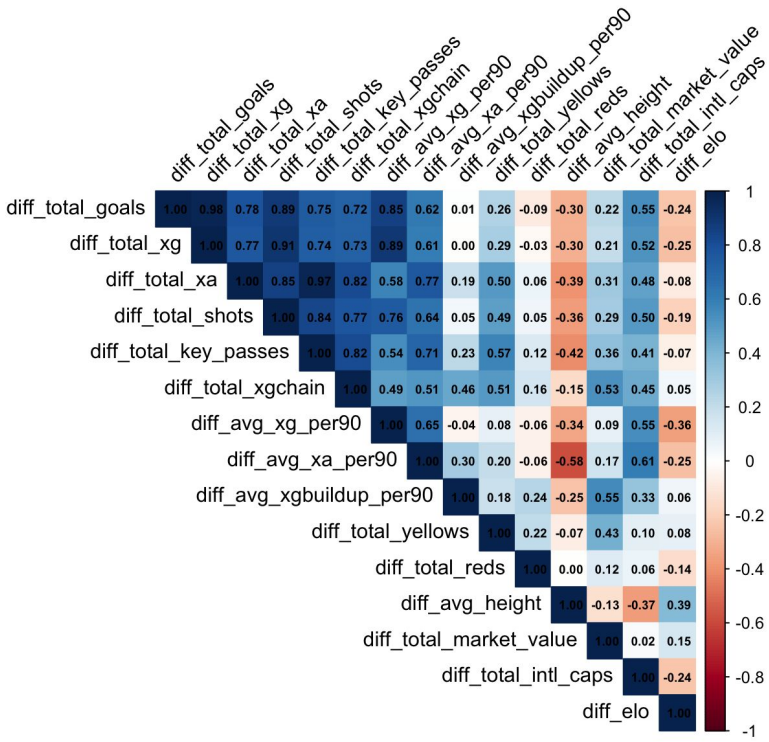
3. Model Preparation & Preprocessing

- Divided the data into an 80% training set (older matches) and a 20% testing set (newer matches up to 2026)
- Centered and scaled all differential numerical features using parameters derived solely from the training data.

N = 916 (for MLR and SVM)

N = 1,672 (for RF and XGBoost)

N = 152 (Recent form baseline for RF and XGBoost)



Methods

- Multiclass Logistic Regression
- Linear SVM
- Random forest (off 200 Monte-Carlo Simulations)
- XGBoost (off 200 Monte-Carlo Simulations)

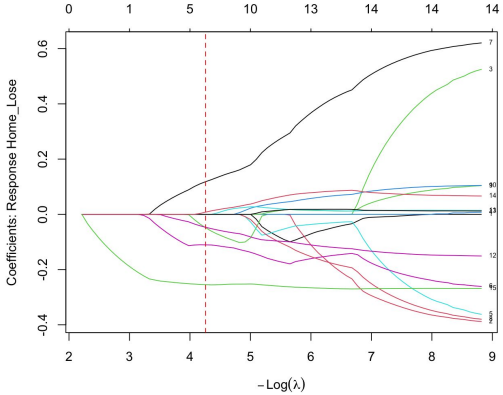
For all of these methods we used class weights due to the imbalance of the match results.

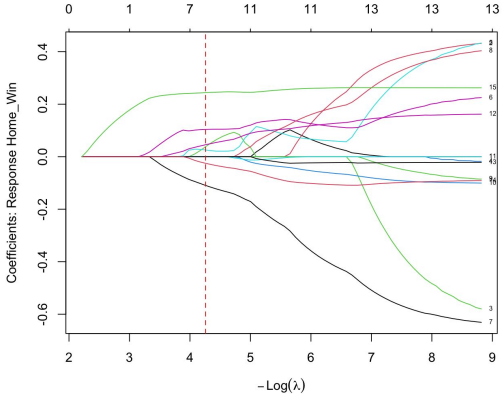
CV was used with Lasso to find a compact LR model and reduce variables.

Also used CV for the cost parameter of the SVM.

Results

LR





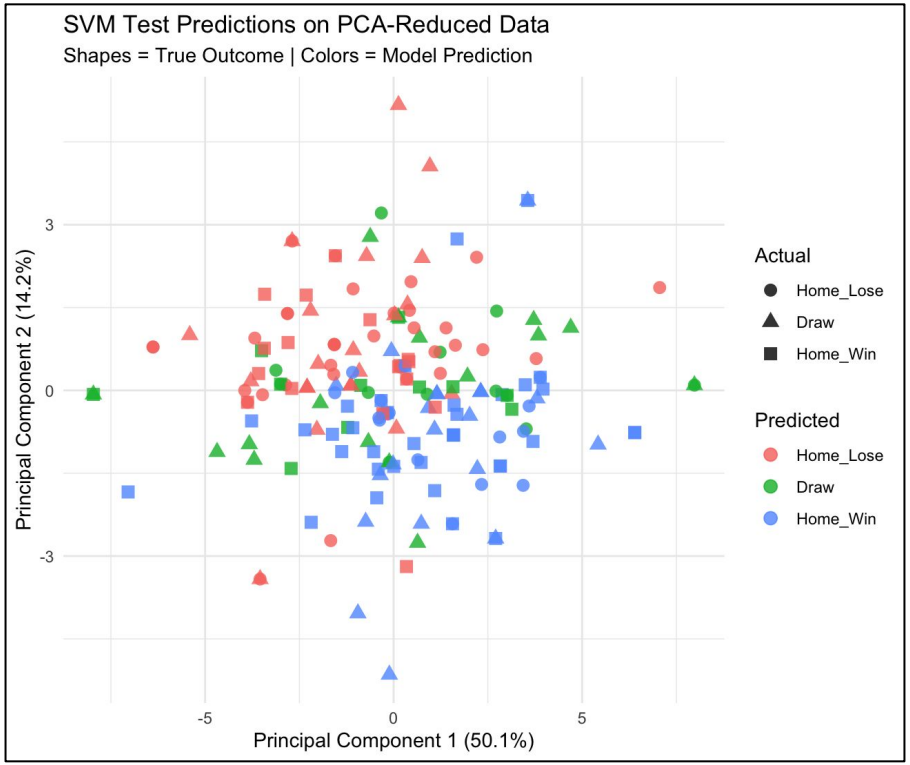

```
$Home_Lose
(Intercept)              -0.017481244
diff_total_xa            -0.048436108
diff_total_xgchain       -0.110646210
diff_avg_xg_per90         0.118190680
diff_avg_height          -0.046867835
diff_total_intl_caps      0.007961881
diff_elo                 -0.254660856

$Draw
(Intercept)               0.0344427

$Home_Win
(Intercept)              -0.01696146
diff_total_xa             0.03808554
diff_total_key_passes     0.02570366
diff_total_xgchain        0.10367296
diff_avg_xg_per90        -0.10947036
diff_avg_height           0.04569252
diff_total_intl_caps     -0.02512454
diff_elo                  0.24395024
```

Test Set Accuracy: 48.91%

	Reference		
Prediction	Home_Lose	Draw	Home_Win
Home_Lose	39	24	15
Draw	9	10	8
Home_Win	15	23	41

	Class	Precision	Recall	F1_Score
1	Class: Home_Lose	0.5000	0.6190	0.5532
2	Class: Draw	0.3704	0.1754	0.2381
3	Class: Home_Win	0.5190	0.6406	0.5734

SVM

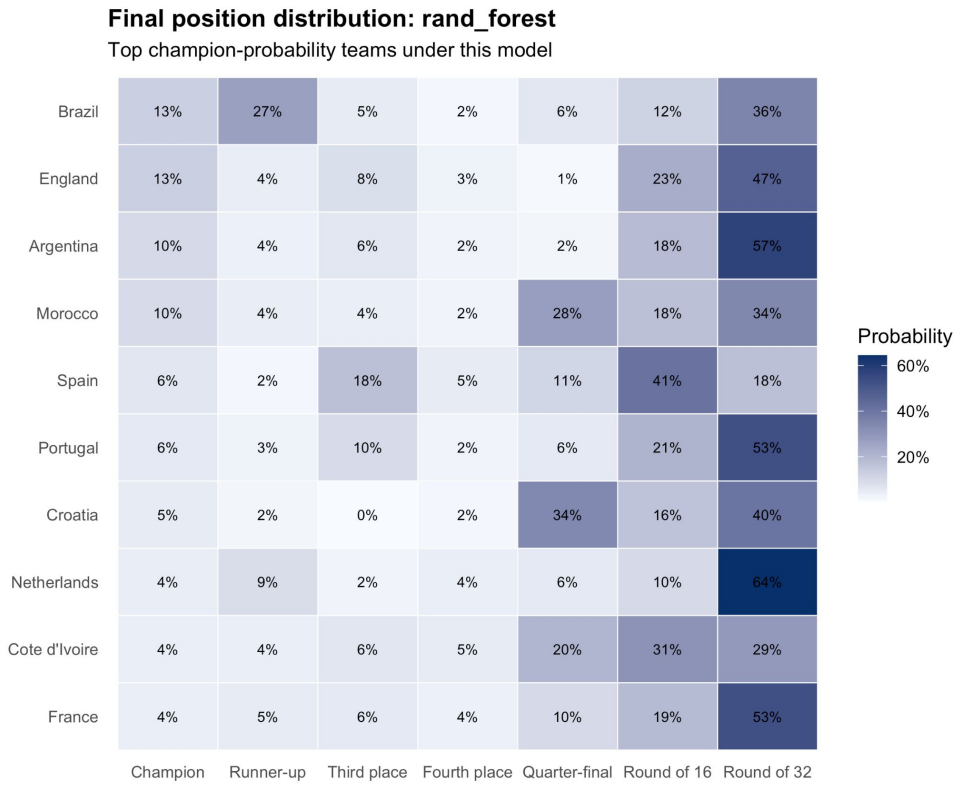


Test Set Accuracy: 47.28%

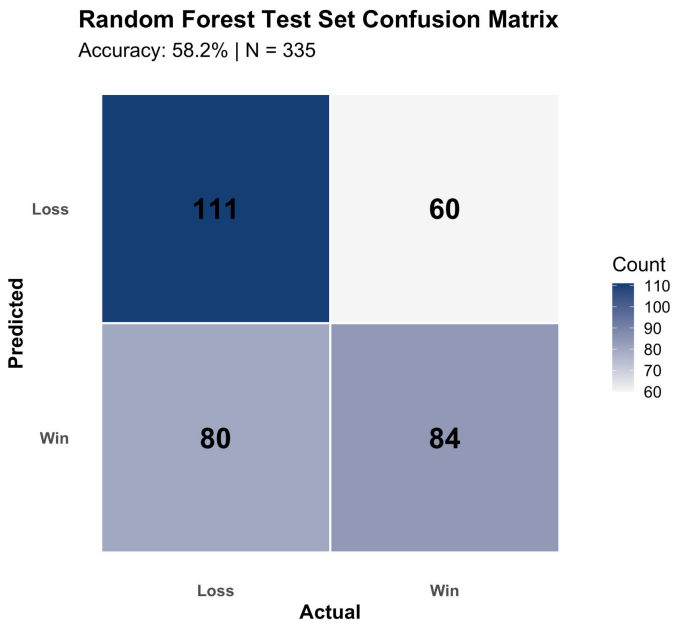
	Actual		
Predicted	Home_Lose	Draw	Home_Win
Home_Lose	35	21	17
Draw	12	16	11
Home_Win	16	20	36

	Precision	Recall	F1
Home_Lose	0.4795	0.5556	0.5147
Draw	0.4103	0.2807	0.3333
Home_Win	0.5000	0.5625	0.5294

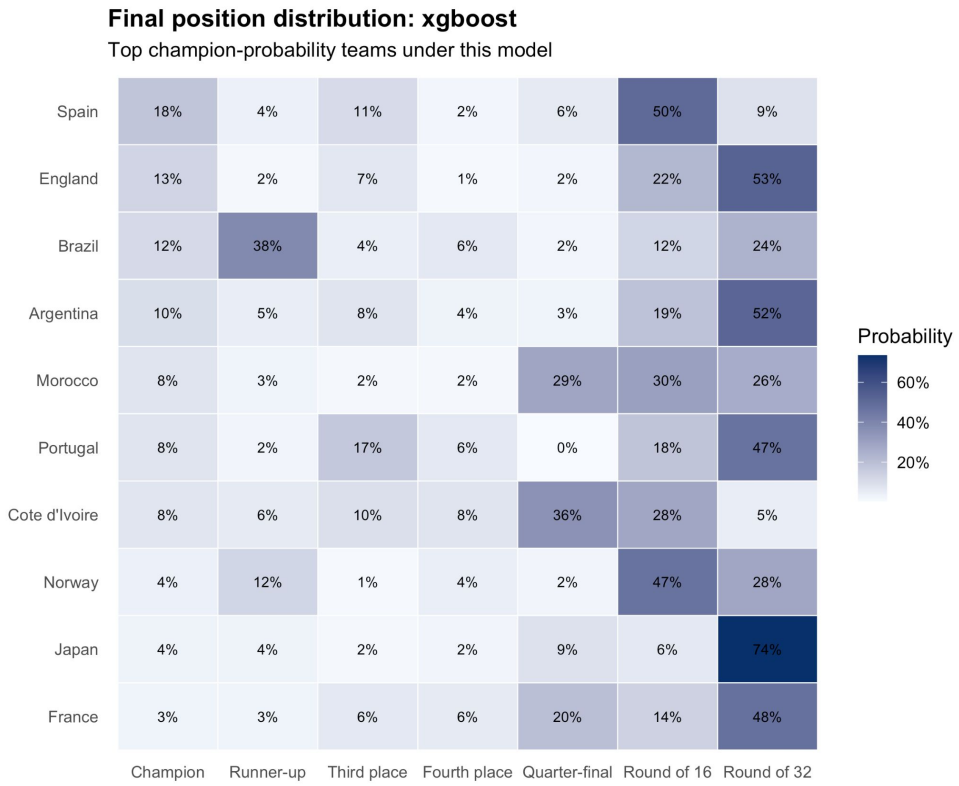
RF



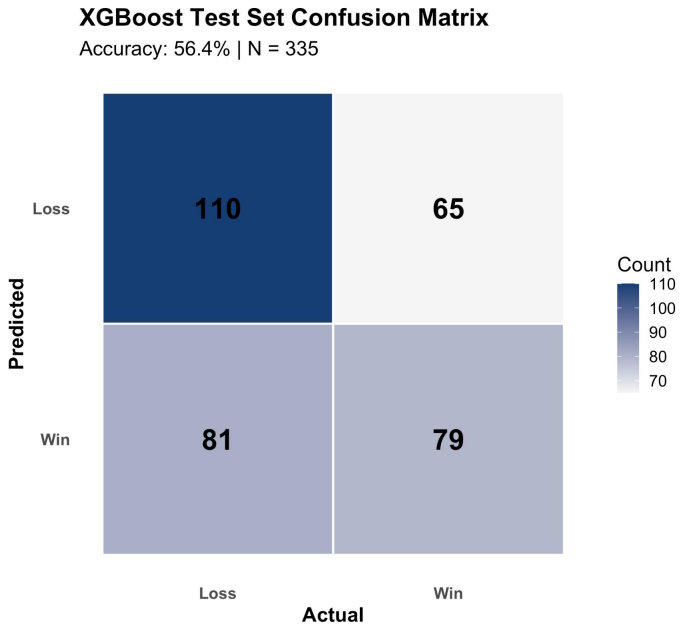
Test Set Accuracy: 58.2%



XGBoost

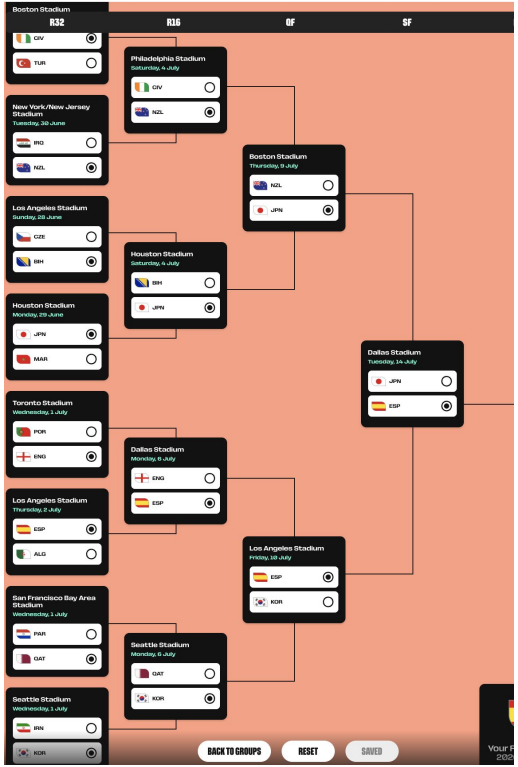


Test Set Accuracy: 56.4%

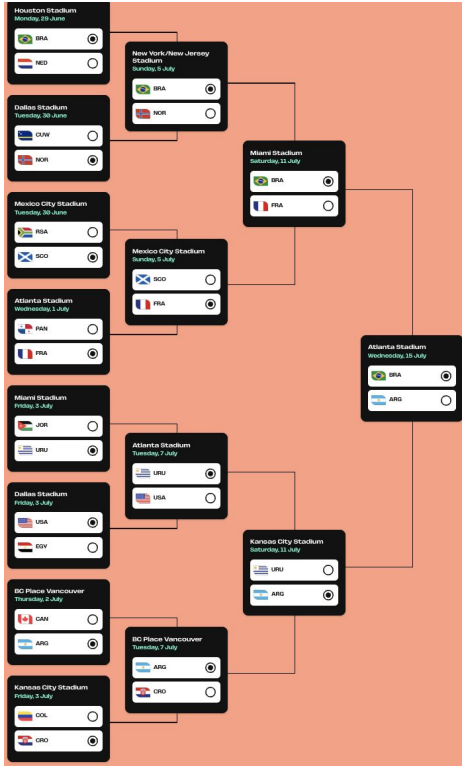


RF Bracket Prediction (Best Model)

Left Side



Right Side



Winner

Conclusions and future work

Conclusions

- Logistic Regression is capable of predicting to a certain degree match results, doing better than random chance
- Draws are the hardest outcome to predict, group-stage draws matter, so expected points is better than forcing a winner
- SVM was not able to find a better boundary than Logistic Regression.
- Obtaining player level statistics is difficult. Club data vs International data.
- LR was better overall not only because it obtained a better accuracy but also because it is more interpretable.
- xG-based squad features are highly predictive.
- Knockout forecast is more realistic when scorelines, extra time, and penalties are included.

Future Work

- Enhance how we obtain player level data (International Data-wise).
- Think of different ways to include player data into a match. Go into more granular detail.
- Incorporate maybe a hybrid scoring model for the Group Stage, and use these models to just simulate knockout rounds.
- Include data from more countries, not necessarily World Cup, to train the model.
- Lock in final squad lists to narrow down total team xG to accurate levels.
- Improve penalty/extra-time calibration.
- Use rolling validation.